Plastic Rods

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
HDPE Rod (Machining Grade)	IS 7328 (High Density Polyethylene Material);IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	ISO 1872-1 (Plastics — Polyethylene (PE) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D6263 (Extruded Rods and Bars Made From Rigid PVC and CPVC – not applicable to HDPE); ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D792 (Density and Specific Gravity of Plastics)	DIN EN ISO 1872-1 (Polyethylene Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Charpy Impact Strength)	UL 94 (where electrical applications apply); FDA 21 CFR 177 (food-contact grades, where applicable); Engineering Plastics Manufacturer Specifications
HDPE Rod (Food Grade)	IS 7328 (High Density Polyethylene Material); IS 10141 (Food Contact Applications); IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	ISO 1872-1 (Plastics — Polyethylene (PE) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 22000 (Food Safety Management Systems, where food contact is applicable); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D4976 (Polyethylene Plastics Molding and Extrusion Materials); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D792 (Density and Specific Gravity of Plastics); ASTM D1894 (Coefficient of Friction of Plastic Film and Sheeting – only where friction testing of the rod/material is relevant)	DIN EN ISO 1872-1 (Polyethylene Moulding and Extrusion Materials); DIN EN ISO 1183-1 (Density of Plastics); DIN EN ISO 1133-1 (Melt Mass Flow Rate of Thermoplastics); DIN EN 1186 (Materials and Articles in Contact with Foodstuffs)	FDA 21 CFR 177; NSF/ANSI 51 (Food Equipment Materials); EU Food Contact Requirements
PP Rod (Industrial)	IS 10910 (Polypropylene Material);IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	ISO 1873-1 (Plastics — Polypropylene (PP) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 15013 (Requirements and tolerances for extruded semi-finished rods of Polypropylene (PP) [2.1]); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D790 (Flexural Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load – where thermal performance is relevant)	DIN EN ISO 1873-1 (Polypropylene Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	UL 94 (Electrical Applications); Engineering Plastics Industry Specifications
PP Rod (Food Contact)	IS 10910 (Polypropylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Food Contact);IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	ISO 1873-1 (Plastics — Polypropylene (PP) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 22000 (Food Safety Management Systems, where applicable); ISO 15013 (Requirements and tolerances for extruded semi-finished rods of Polypropylene (PP) [2.1]); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D4101 (Polypropylene Injection and Extrusion Materials); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D790 (Flexural Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load – where relevant); ASTM D6400 (Compostable Plastics – only if the product is specifically claimed as compostable)	DIN EN ISO 1873-1 (Polypropylene Moulding and Extrusion Materials); DIN EN ISO 1183-1 (Density of Plastics); DIN EN ISO 1133-1 (Melt Mass Flow Rate); DIN EN 1186 (Materials and Articles in Contact with Foodstuffs)	FDA 21 CFR 177; NSF/ANSI 51; EU Food Contact Requirements
PVC Rod (Rigid)	IS 10151 (Polyvinyl Chloride (PVC) and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Food Contact);IS 14193; IS	ISO 1163-1 (Plastics — Unplasticized Poly(vinyl chloride) (PVC-U) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D6263 (Extruded Rods and Bars Made From Rigid PVC and CPVC); ASTM D1784 (Rigid PVC and CPVC Compounds – material classification); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural	DIN EN ISO 21306-1 (PVC-U Moulding and Extrusion Materials); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	UL 94 (Flammability Classification); NSF/ANSI 61 (Water Applications, where applicable)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	14434; IS 11297; IS 1998; IS 13360		Load – where relevant)		
CPVC Rod	IS 15778 (Chlorinated Polyvinyl Chloride (CPVC) Compounds for Hot and Cold Water Applications);IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	ISO 11613 (Plastics — Chlorinated Poly(vinyl chloride) (PVC-C) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D6263 (Extruded Rods and Bars Made From Rigid PVC and CPVC); ASTM D1784 (Rigid PVC and CPVC Compounds – material classification); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load – where relevant)	DIN EN ISO 1163-1 (PVC Moulding and Extrusion Materials); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	NSF/ANSI 61; UL 94; Chemical Process Industry Specifications
Nylon Rod (PA6); Nylon Rod (PA66)	IS 1843;IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	Nylon Rod (PA66);ISO 1874-1 (Plastics — Polyamide (PA) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 16396-2 (Specimen preparation and material verification layouts for Polyamide (Nylon) rods [2.1]); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D4066 (Nylon Injection and Extrusion Materials); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D790 (Flexural Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load – where relevant)	DIN EN ISO 16396-1 (Polyamide Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Charpy Impact Strength)	UL 94 (Flammability Classification); FDA 21 CFR 177 (Food Contact Grades); Engineering Plastics Industry Specifications
Acetal (POM) Rod; Delrin Rod (POM-H)	IS 9845 (Migration Limits for Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	Delrin Rod (POM-H);ISO 9988-1 (Plastics — Polyoxymethylene (POM) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 15526-2 (Extruded semi-finished profiles and solid rods of Polyoxymethylene (POM)); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D6100 (Extruded, Compression Molded and Injection Molded Polyoxymethylene Shapes); ASTM D6778 (Polyoxymethylene Molding and Extrusion Materials – material classification); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load); ASTM D3702 (Wear Rate and Coefficient of Friction of Self-Lubricating Materials – where bearing/wear application is relevant)	DIN EN ISO 9988-1 (Polyoxymethylene (POM) Moulding and Extrusion Materials); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Charpy Impact Strength)	FDA 21 CFR 177 (Food Contact Applications); NSF/ANSI 61 (Water Applications); UL 94
PTFE Rod	IS 4894; IS 14434:1997 (Polytetrafluoroethylene (PTFE) Rods — Specification);IS 14193; IS 11297; IS 1998; IS 13360	ISO 12086-1 (Plastics — Fluoropolymer Dispersions and Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 13000-2 (Preparation of test specimens and determination of properties for processed PTFE semi-finished rods); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D1710 (Extruded PTFE Rod, Heavy-Walled Tubing and Basic Shapes); ASTM D4894 (PTFE Granular Molding and Ram Extrusion Materials – resin/material); ASTM D4745 (Filled PTFE Molding and Extrusion Materials – where filled PTFE is used); ASTM D638 (Tensile Properties of Plastics); ASTM D3702 (Wear Rate and Coefficient of Friction of Self-Lubricating Materials – where applicable)	DIN EN ISO 13000-1 (Polytetrafluoroethylene (PTFE) Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 12086-1 (Fluoropolymer Materials); DIN EN ISO 178 (Flexural Properties)	FDA 21 CFR 177 (Food Contact PTFE); USP Class VI (Medical Applications); UL 94
PVDF Rod	IS 9845 (Migration Limits for Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 14434:1997 (Polytetrafluoroethylene	ISO 12086-1 (Plastics — Fluoropolymer Materials — Designation System and Basis for Specifications); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D6713 (Extruded and Compression Molded Shapes Made From PVDF); ASTM D3222 (Unmodified PVDF Molding, Extrusion and Coating Materials – resin/material); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of	DIN EN ISO 12086-1 (Fluoropolymer Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177 (Food Contact Applications); NSF/ANSI 61; Semiconductor Industry Material Specifications

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	(PTFE) Rods — Specification);IS 14193; IS 11297; IS 1998; IS 13360		Plastics); ASTM D543 (Resistance of Plastics to Chemical Reagents – where chemical service is relevant)		
UHMWPE Rod	IS 10910 (Polyethylene and its Copolymers for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Food Contact);IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	ISO 11542-1 (Plastics — Ultra-high Molecular Mass Polyethylene (PE-UHMW) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D6712 (UHMW-PE Solid Plastic Shapes); ASTM D4020 (UHMW-PE Molding and Extrusion Materials – resin/material); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load – where relevant)	DIN EN ISO 11542-1 (Ultra-High Molecular Weight Polyethylene Moulding Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177; USP Class VI (Medical Applications); NSF/ANSI 61
PEEK Rod	IS 9845 (Migration Limits for Plastics in Contact with Foodstuffs) (Food Contact Applications); IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	ISO 16354 (Plastics — Polyetheretherketone (PEEK) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D6262 (Extruded, Compression Molded, and Injection Molded Basic Shapes of PAEK); ASTM D8033 (PEEK Molding and Extrusion Materials – material classification); ASTM D638 (Tensile Properties of Plastics); ASTM D790 (Flexural Properties of Unreinforced and Reinforced Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load – where relevant)	DIN EN ISO 21318-1 (Polyetheretherketone Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	FDA 21 CFR 177; USP Class VI; UL 94
ABS Rod	IS 17077(ABS molding and extrusion materials); IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	ISO 2580-1 (Plastics — Acrylonitrile-butadiene-styrene (ABS) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D4673 (ABS Plastics and Alloys Molding and Extrusion Materials); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load – where relevant)	DIN EN ISO 2580-1 (ABS Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	UL 94; UL 746C (Polymeric Materials for Electrical Equipment); Engineering Plastics Industry Specifications
Acrylic (PMMA) Rod	IS 10139 (Acrylic Resins for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Food Contact); IS 14193:1994 (Acrylic Plastic Rods and Tubes for General Engineering Purposes — Specification);IS 14434; IS 11297; IS 1998; IS 13360	ISO 8257-1 (Plastics — Poly(methyl methacrylate) (PMMA) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 24022-2 (Performance properties and testing limits for Polymethyl Methacrylate (PMMA) solid rods [2.1]); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D788 (PMMA Molding and Extrusion Materials); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D1003 (Haze and Luminous Transmittance of Transparent Plastics – where optical transparency is relevant)	DIN EN ISO 8257-1 (PMMA Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	UL 94; UL 746C; Building Material Fire Classification Requirements
Polycarbonate Rod	IS 14434 (Polycarbonate Sheets for Roofing and Cladding) (Polycarbonate Applications); IS 14193; IS 11297; IS 1998; IS 13360	ISO 21305-1 (Plastics — Polycarbonate (PC) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 7391-2 (Structural classification of semi-finished Polycarbonate (PC) extrusion bars [2.1]); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D6098 (Extruded and Compression Molded Shapes Made From Polycarbonate); ASTM D3935 (Polycarbonate Unfilled and Reinforced Material – material classification); ASTM D638 (Tensile Properties of Plastics); ASTM D256 (Izod Impact Resistance of Plastics); ASTM D648 (Heat Deflection Temperature of Plastics)	DIN EN ISO 21305-1 (Polycarbonate Moulding and Extrusion Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 179-1 (Impact Strength)	UL 94; UL 746C; NSF/ANSI 61 (where applicable)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
PET Rod; PETG Rod	IS 12252 (Polyethylene Terephthalate (PET) for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water) (Food Contact);IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	PETG Rod;ISO 7792-1 (Plastics — Poly(ethylene terephthalate) (PET) Moulding and Extrusion Materials — Designation System and Basis for Specifications); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D6261 (Extruded and Compression Molded Basic Shapes Made From Thermoplastic Polyester); ASTM D4603 (Inherent Viscosity of PET – applicable to PET material); ASTM D638 (Tensile Properties of Plastics); ASTM D790 (Flexural Properties of Plastics); ASTM D648 (Deflection Temperature of Plastics Under Flexural Load – where relevant)	DIN EN ISO 7792-1 (Thermoplastic Polyester Moulding Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN ISO 11357 (Thermal Properties by DSC)	FDA 21 CFR 177; EU Food Contact Requirements; UL 94 (where applicable)
ESD Plastic Rod	IS 15748 (Electrostatic Discharge Protection Applications); IS 14193; IS 14434; IS 11297; IS 1998; IS 13360	ISO 11469 (Plastics — Generic Identification and Marking of Plastics Products); IEC 61340-5-1 (Electrostatics — Protection of Electronic Devices from Electrostatic Phenomena); ISO 604 (General test method for determining compressive properties of plastic rod elements)	ASTM D257 (DC Resistance or Conductance of Insulating Materials); ASTM D638 (Tensile Properties of Plastics); ASTM E595 (Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in Vacuum Environment – only for applicable electronic/space/vacuum applications); ASTM D256 (Izod Impact Resistance of Plastics – where relevant)	DIN EN 61340-5-1 (Electrostatics — Protection of Electronic Devices from Electrostatic Phenomena); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 178 (Flexural Properties); DIN EN 61340-2-3 (Methods for Determining Resistance and Resistivity of Flat Materials)	IEC 61340 Series (Electrostatic Control); ANSI/ESD S20.20 (ESD Control Programme Requirements); UL 94

Source field: Plastic and Plastic Products